

Design and Comparison of LCL Filters in STATCOM

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Abstract—The LCL filter is indispensable for grid-connected inverters, as it masterfully mitigates harmonic distortion in the output current. By effectively attenuating high-frequency harmonics, the filter not only ensures compliance with stringent grid standards, but also significantly enhances overall power quality, thus safeguarding against risks such as overheating and equipment damage. This paper investigates the performance of an LCL filter integrated with a 50 kVAR STATCOM, comparing two design methods based on their inductance values. The simulation results from the proposed design reveal a remarkable improvement in waveform quality, producing a smooth sinusoidal output with minimal harmonic distortion.

Index Terms—STATCOM, LCL filter, Damping, Resonance

I. INTRODUCTION

A STATCOM (Static Synchronous Compensator) is an advanced power electronic device used for dynamic reactive power compensation in electrical grids [1]. It consists of a Voltage Source Inverter (VSI) that is connected to the grid through a filter, typically an L or LCL filter, to ensure smooth operation and enhance power quality, as shown in Fig.1. The primary function of a STATCOM is to regulate and balance the reactive power in the system. In doing so, it helps maintain voltage stability, improve the power factor, and support the overall performance of the grid, particularly during fluctuating load conditions or disturbances.

STATCOMs can either supply or absorb reactive power as needed in real-time, ensuring effective voltage regulation. Unlike traditional compensators, which are slower to react, a STATCOM, which uses a voltage source converter, responds almost immediately to changes in the grid [2]. This fast response is crucial for improving the reliability of the system and alleviating issues such as voltage fluctuations, especially in grids that experience variable loads or integrate with renewable energy sources.

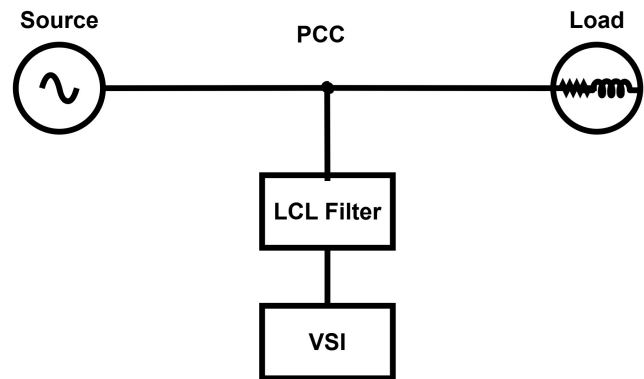


Fig. 1. Single line diagram of STATCOM

The integration of an LCL filter with the STATCOM further enhances its functionality by reducing harmonic distortion and improving overall system efficiency.

LCL filters are designed to mitigate the harmonic currents produced by grid-connected inverters, improving the quality of the power injected into the grid. These filters consist of two inductors (L) and a capacitor (C), which together form a third-order low-pass filter. The inductors in the filter act as impedance to high-frequency switching currents, which are the primary source of harmonics in inverter output. By blocking or attenuating these high-frequency components, the inductors reduce the harmonic content of the current flowing from the inverter into the grid. The capacitor, placed between the two inductors, provides a low impedance path to the high-frequency switching components, allowing them to be dissipated, effectively reducing harmonic distortion. As a result, LCL filters are highly effective at eliminating unwanted high-frequency noise while maintaining the integrity of the AC waveform, ensuring that the energy injected into the grid is

of good quality.

The design of LCL filters is more complex than traditional L filters, which consist only of inductors. While L filters can also reduce harmonic distortion to some extent, they do so less effectively because they offer less impedance at high frequencies compared to LCL filters. However, the effectiveness of LCL filters is not without its challenges. A key issue in their design is the potential for resonance, which can occur at certain frequencies where the filter's impedance is low. Resonance can lead to a significant increase in current at those frequencies, creating oscillations that can destabilize the system and degrade power quality. This is particularly problematic when the resonant frequency coincides with the inverter's switching frequency or other dominant frequencies in the grid. To avoid this, careful attention must be paid to the placement of the resonance frequency in the filter design, ideally between ten times the line frequency and half of the switching frequency. This range helps to ensure that the filter attenuates high-frequency noise without inadvertently introducing resonance that could compromise system stability.

To manage resonance and maintain the stability of the filter, damping techniques are used. These methods can be broadly classified into passive and active approaches. Active damping involves adding control elements—such as feedback loops or additional circuit components—to the filter design to counteract the resonant frequencies [3]. This approach enhances system stability and ensures better control over resonance, but it introduces complexity, potentially increasing the cost and reducing control bandwidth. Passive damping, on the other hand, relies on the inherent properties of the filter circuit, such as parasitic resistances in the inductors and capacitors, to dissipate energy and reduce resonance. While passive damping is simpler and cost-effective, it may introduce small losses in power efficiency [4]. However, these losses are typically minimal and do not significantly impact the overall performance of the system.

The design choice between active and passive damping strategies depends on several factors, including the specific application, cost constraints, and the desired level of efficiency [5]. In practice, passive damping is often preferred for its simplicity and effectiveness in mitigating resonance [6], but active damping is considered when the system requires higher precision in controlling resonant behaviour. Passive and active damping methods for resonance suppression in LCL filters used in active power filter (APF) applications are compared in [7]. As research in the field of grid-connected inverters continues to advance, new filter designs and control strategies are emerging, further optimizing inverter performance and enhancing the stability and reliability of renewable energy-integrated grids. These innovations are crucial in ensuring that the growing share of renewable energy sources, such as solar and wind, can be reliably integrated into the power grid without compromising power quality or system stability. There are design procedures for LCL filters used in grid-connected voltage-source inverters (VSI) to reduce inductance values and cost while maintaining the same filtering effectiveness [8]–

[12]. Common-mode voltage (CMV) mitigation techniques in electric drive systems, with a focus on inverter topology variants and modulation strategies, and an optimized LCL filter design for harmonic attenuation in a DFIG-based wind energy system, demonstrating its effectiveness in reducing harmonic distortion and improving reactive power compensation explained in [13], [14]. This paper explains a LCL filter with damping in a 50kVAR STATCOM.

II. DAMPING METHODS FOR LCL FILTER

LCL filters are widely used in power electronic systems to mitigate harmonics and improve power quality. However, these filters can experience resonance at specific frequencies, which can degrade their performance and potentially lead to system instability. To address this issue, various damping techniques are employed to suppress these resonances and ensure effective harmonic attenuation.

A. Series Resistor Damping

One of the simplest and most commonly used methods is series resistor damping, where a resistor is placed in series with the filter capacitor. This resistor helps dissipate oscillatory energy, thereby reducing resonance effects. However, a major drawback of this method is its inherent power loss, particularly at higher frequencies, where it can lead to reduced system efficiency. Despite this limitation, its simplicity and ease of implementation make it a practical choice in many applications.

B. Shunt RC Damping

To overcome the high-power losses associated with series resistor damping, the shunt RC damper is used. This method involves placing a resistor and a capacitor in parallel with the filter capacitor. The capacitor provides a bypass path for high-frequency components, while the resistor dissipates oscillations effectively. Compared to series resistor damping, this approach achieves better high-frequency attenuation while significantly reducing power losses. As a result, shunt RC damping is often preferred in applications where efficiency is a key consideration.

C. Shunt RLC Damping

For even better resonance suppression and lower power dissipation, the shunt RLC damper can be implemented. In this method, an additional inductor is placed in parallel with the resistor and capacitor, forming a second-order damping circuit. The inductor bypasses the fundamental current, ensuring that the resistor primarily dissipates oscillations at higher frequencies. This results in efficient resonance suppression with minimal impact on fundamental frequency operation, making it a more optimized solution for high-performance power electronics.

D. Series RLC Damping

Another advanced damping technique is series RLC damping, where an RLC circuit is placed in series with the filter capacitor. Unlike the previous methods, this approach provides greater control over resonance by tailoring the impedance characteristics of the circuit. The resistor is strategically bypassed at low and high frequencies, ensuring that damping is primarily active around the resonance frequency. While this method is effective, its increased complexity and the need for precise tuning make it more challenging to implement in practical systems.

E. Trap Filters for Harmonic Attenuation

Trap filters provide an alternative method for mitigating resonance and harmonics. These filters are designed to exhibit very low impedance at specific switching harmonic frequencies, thereby effectively filtering out unwanted frequency components. Compared to traditional LCL filters, trap filters are often more compact and can be integrated with shunt RC dampers for enhanced high-frequency attenuation. However, their effectiveness diminishes above the tuning frequency, limiting their application to specific frequency ranges.

F. 2-Traps Filter with Shunt RC Damping

To further optimize filter size and improve performance, the 2-traps filter with shunt RC damper can be used. This method employs two tuned traps, each targeting different harmonic frequencies, leading to enhanced harmonic attenuation while minimizing overall filter volume. By incorporating a shunt RC damper, high-frequency attenuation is further improved, making it an attractive choice for applications where space constraints and high filtering efficiency are critical.

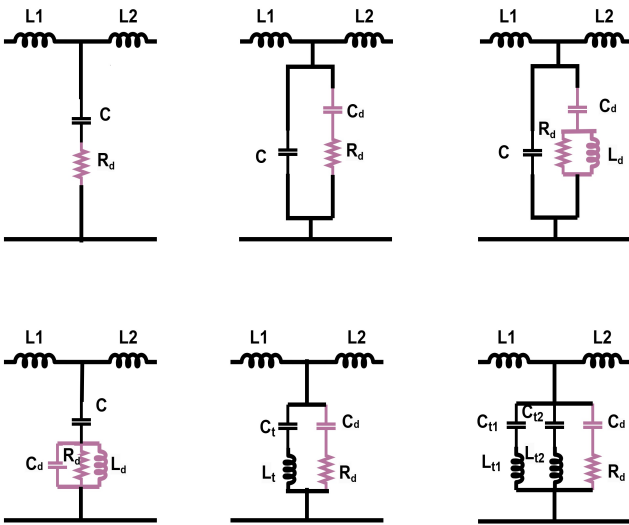


Fig. 2. Damping circuits for LCL Filter

III. DESIGN OF LCL FILTER

A. Transfer Function

The transfer function of the grid-connected LCL filter describes the relationship between the input voltage V_i and the grid current I_g , taking into account the dynamics of the filter components. As shown in Fig. 3, the basic structure of an LCL filter consists of two inductors L_1 and L_2 , and a capacitor C , which form a resonant circuit that attenuates high-frequency harmonics while allowing the fundamental frequency to pass through.

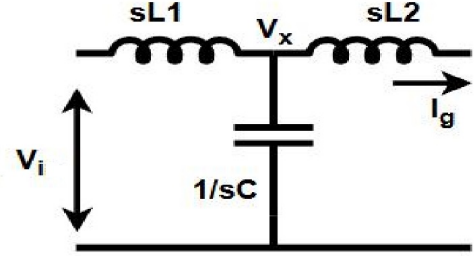


Fig. 3. Basic Structure of an LCL Filter

Starting with the initial voltage-current relationship across the components, equation (1) is derived from Kirchhoff's voltage law and the impedance properties of the inductor and capacitor in the Laplace domain. This equation captures the interaction between the input voltage V_i , the voltage across the inductor-capacitor network (V_x), and the grid current I_g :

$$\frac{V_i - V_x}{sL_1} = I_g + \frac{V_x}{\frac{1}{sC}} \quad (1)$$

$$V_x = I_g sL_2 \quad (2)$$

By rearranging equation (1) and using the expression for V_x from equation (2) ($V_x = I_g sL_2$), the transfer function simplifies further. This results in a third-order equation that describes the system's dynamics:

$$\frac{I_g}{V_i} = \frac{1}{s^3 L_1 L_2 C + s(L_1 + L_2)} \quad (3)$$

$$\text{Let } L_1 + L_2 = L \quad (4)$$

Next, on simplifying the terms, the parallel inductance L_p is defined as:

$$L_p = \frac{L_1 L_2}{L_1 + L_2} \quad (5)$$

Substituting these into equation (3) results in the final transfer function of the LCL filter:

$$\frac{I_g}{V_i} = \frac{1}{sL(s^2 L_p C + 1)}$$

This transfer function characterizes the frequency-domain behaviour of the LCL filter, where the denominator reflects

the combined inductive and capacitive impedance of the filter. The transfer function is crucial for analysing the system's resonance frequency, bandwidth, and filtering performance. It also provides insights into how the filter will attenuate higher-order harmonics and how the system behaves at different frequencies, making it an essential tool for designing and optimizing LCL filters in grid-connected applications.

B. LCL Filter with damping

A good high-frequency attenuation and very low damping loss can be obtained by the second-order damped filter (RLC circuit in parallel with the filter capacitor) as illustrated in Fig.4, since the fundamental current in the resistor is bypassed by the additional damping inductance L_d .

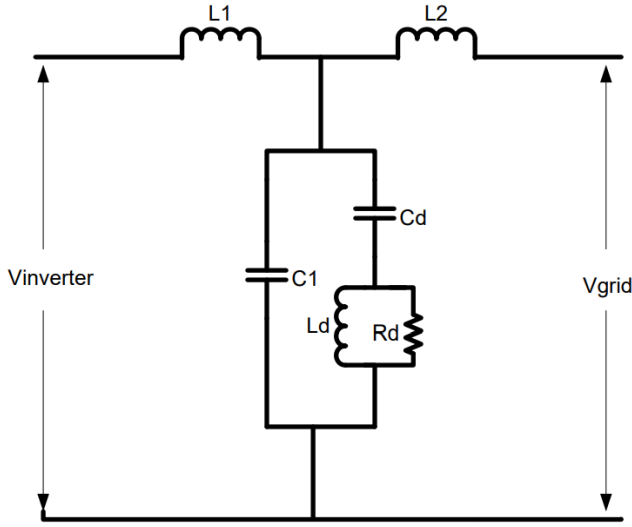


Fig. 4. LCL Filter with damping

- S: Apparent Power
- Q: Reactive Power
- L1: Inverter side inductance
- L2: Grid side inductance
- C1: filter capacitance
- I_{inv} : Inverter current
- f_{sw} : Switching Frequency(25kHz)
- f_{res} : Resonant Frequency(2.5kHz)

1) **Method 1:** In this method, the maximum value of capacitor is calculated by considering the reactive power as 5% of the rated KVA (50KVA). The filter capacitance and damping capacitance will be taken as the same and is calculated as half of the maximum capacitance(C). The value of inductance is given by equation (9). Finally, the damping inductance and resistance are calculated.

STEP 1: FINDING C

Reactive power Q is given by:

$$Q = 5\% \text{ of } S \quad (6)$$

Also,

$$Q = \frac{v^2}{\frac{1}{2\pi fC}} \quad (7)$$

Where:

$$v = 415 \text{ V}, \quad f = 50 \text{ Hz}$$

Calculating Q :

$$Q = 0.05 \times 50000 = 2500 \text{ VA}$$

Substituting the above values in equation (7):

$$C = \frac{Q}{2\pi f v^2} = \frac{2500}{2\pi \times 50 \times 415^2} \approx 46.23 \mu F$$

Thus,

$$C_d = \frac{C}{2} = 23.11 \mu F$$

$$C_1 = \frac{C}{2} = 23.11 \mu F$$

STEP 2: FINDING L

The resonant angular frequency:

$$\omega_{res} = \frac{1}{\sqrt{CL_P}} \quad (8)$$

And:

$$\omega_{res} = 2\pi f_{res}$$

From (4) and (5), if L_1 and L_2 are equal

$$L_P = \frac{L}{4}$$

Then:

$$L = \frac{4}{(2\pi f_{res})^2 C} \quad (9)$$

Substituting $f_{res} = 2500 \text{ Hz}$ and $C = 46.23 \mu F$:

$$L = \frac{4}{(2\pi \times 2500)^2 \times 46.23 \times 10^{-6}} \approx 350 \mu H$$

Now, $L = L_1 + L_2$

Thus:

$$L_1 = L_2 = \frac{350 \mu H}{2} = 175 \mu H$$

STEP 3: FINDING THE VALUE OF DAMPING INDUCTOR L_d

$$L_d = L = 350 \mu H$$

STEP 4: FINDING THE VALUE OF DAMPING RESISTOR R_d

$$R_d = \sqrt{\frac{L}{C}} = \sqrt{\frac{350 \times 10^{-6}}{46.23 \times 10^{-6}}} \approx 2.75 \Omega$$

2) **Method 2:** In this method, instead of taking the reactive power, considered the voltage across inductor as 10% of phase voltage. Then, value of inductor is calculated by equation (10) and (11). Finally, damping values are calculated as similar to method 1.

STEP 1: TAKING VOLTAGE ACROSS INDUCTOR

The voltage across the inductor is given by:

$$V_L = 10\% \times V_{grid} \quad (10)$$

STEP 2: FINDING L

The inverter current I_{inv} can be calculated as:

$$I_{inv} = \frac{50 \text{ kVA}}{\sqrt{3} \times 415} \approx 70 \text{ A}$$

Using the relationship between voltage, current, and inductance:

$$V_L = 2\pi f L I_{inv} \quad (11)$$

From equations (10) and (11), we have:

$$L = \frac{0.1 V_{grid}}{2\pi f I_{inv}} = \frac{0.1 \times 240}{2\pi \times 50 \times 70} \approx 1 \text{ mH}$$

But, $L = L_1 + L_2$

Thus:

$$L_1 = L_2 = \frac{1 \text{ mH}}{2} = 500 \mu\text{H}$$

STEP 3: FINDING C

From the relationship in (9):

$$C = \frac{4}{(2\pi f_{res})^2 L} \quad (12)$$

Substituting the values:

$$C = \frac{4}{(2\pi \times 2500)^2 \times 1 \times 10^{-3}} \approx 16 \mu\text{F}$$

Thus,

$$C_d = \frac{C}{2} = 8 \mu\text{F}$$

$$C_1 = \frac{C}{2} = 8 \mu\text{F}$$

STEP 4: FINDING THE VALUE OF DAMPING INDUCTOR L_d

$$L_d = L = 1 \text{ mH}$$

STEP 5: FINDING THE VALUE OF DAMPING RESISTOR R_d

$$R_d = \sqrt{\frac{L}{C}} = \sqrt{\frac{1 \times 10^{-3}}{16 \times 10^{-6}}} \approx 7.7 \Omega$$

TABLE I
COMPARISON OF METHOD 1 & METHOD 2

Parameter	Method 1	Method 2
Total Inductance (L)	350 μH	1 mH
$L_1 = L_2 = \frac{L}{2}$	175 μH	500 μH
Capacitance	46.23 μF	16 μF
Damping Inductor (L_d)	350 μH	1 mH
Damping Resistor (R_d)	2.75 Ω	7.7 Ω
Damping Capacitor (C_d)	23.11 μF	8 μF

IV. COMPARISON

Table I represent a comparison of the inductance values of an LCL filter used in conjunction with a 50 kVAR STATCOM. In Method 1, the total inductance is calculated to be 350 microhenries (μH), whereas Method 2 results in a higher total inductance of 1 millihenry (mH).

V. SIMULATION AND RESULTS

Fig.5 shows the simulation circuit of the STATCOM integrated with the LCL filter, highlighting its role in improving voltage regulation and reducing harmonic distortion in the power system. Fig.6 shows the THD analysis of the inverter current that indicates a significant improvement in power quality after filtering. Before filtering, the THD was 8.35%, which was relatively high due to the presence of switching harmonics. After implementing the filter, the THD reduced to 2.65%, demonstrating the effectiveness of the filtering technique in suppressing harmonic distortions. This improvement aligns well with the IEEE 519 standards, which recommend maintaining a current THD below 5% for reliable power system operation. The reduction in harmonics improves the quality of inverter output, leading to lower power losses, reduced heating in connected equipment, and improved overall system efficiency. This result validates the successful implementation of the filtering approach in ensuring better power quality. Fig.7 shows the inverter current waveform with and without the filter, demonstrating the significant improvement in voltage quality and harmonic reduction when the filter is employed. Upon passing through the LCL filter, the output waveform transforms into a smooth sine wave. This dramatic improvement demonstrates the filter's capability to significantly attenuate the harmonics present in the original signal. Sharp peaks and irregularities are replaced by a clean continuous waveform, indicating that the filter has effectively reduced the harmonic content to acceptable levels.

System uncertainties, such as parameter variations and grid disturbances, can significantly impact the performance of an LCL filter integrated with a STATCOM. Variations in inductance, capacitance, or grid impedance can shift the filter's resonance, reducing harmonic attenuation and potentially causing instability. These parameter deviations alter the impedance characteristics of the filter, affecting its ability to suppress unwanted harmonics and maintain a stable operation. Disturbances like voltage sags, transients, or faults can further overwhelm the filter's capacity, leading to poor reactive power compensation, increased total harmonic distortion (THD), and

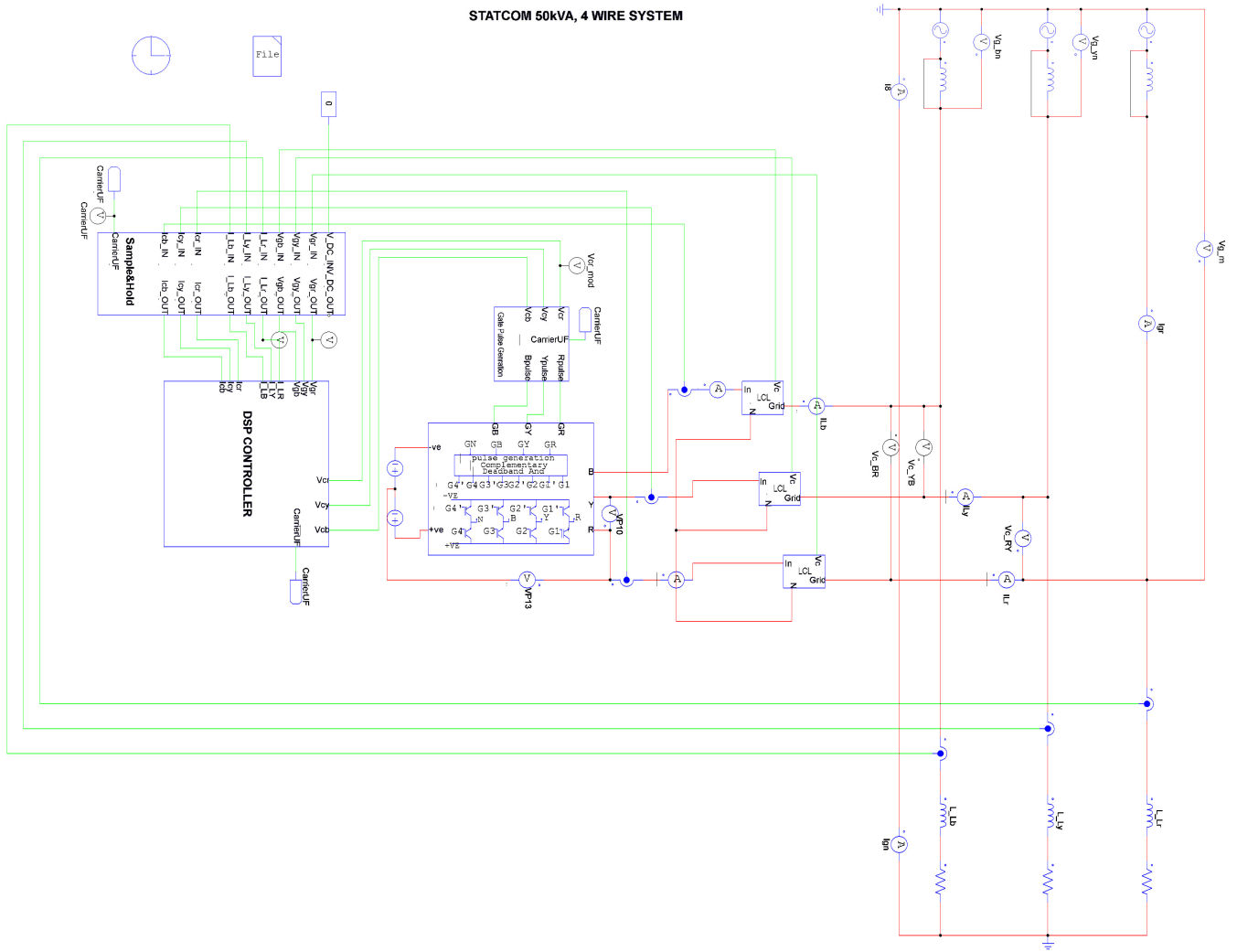


Fig. 5. Simulation circuit of STATCOM with LCL filter

potential system instability. In practical applications, grid conditions are rarely ideal, and fluctuations in grid impedance due to changing load conditions or network reconfigurations can alter the LCL filter's effectiveness, making it essential to design the filter with robustness against such variations.

The LCL filter's design, which combines inductive and capacitive elements to create a resonant circuit, allows it to selectively attenuate specific harmonics while maintaining the integrity of the fundamental frequency. This selective filtering capability is crucial in mitigating high-frequency switching harmonics generated by the STATCOM's power electronic converters, ensuring smooth and stable grid integration. However, under uncertain conditions, such as changes in grid strength or fluctuations in operating conditions, the resonance of the LCL filter may shift, leading to undesired amplification of certain frequency components rather than suppression. To counteract these effects, careful tuning of the filter parameters, adaptive control strategies, or active damping techniques are often employed to enhance system robustness. Ensuring power

quality and compliance with industry standards requires an LCL filter that not only provides effective harmonic attenuation but also adapts to variations in system parameters and grid disturbances. Without proper design considerations, including resonance damping, stability analysis, and adaptive control, the filter may fail to perform optimally in real-world conditions. Therefore, addressing these uncertainties through advanced control techniques, robust component selection, and real-time monitoring is critical for maintaining STATCOM performance, ensuring reliable reactive power compensation, and enhancing the overall stability of the power system.

TABLE II
THD ANALYSIS OF INVERTER CURRENT

Fundamental Frequency	50 Hz
Inverter Current (Before Filtering)	0.0835
Inverter Current (After Filtering)	0.0265

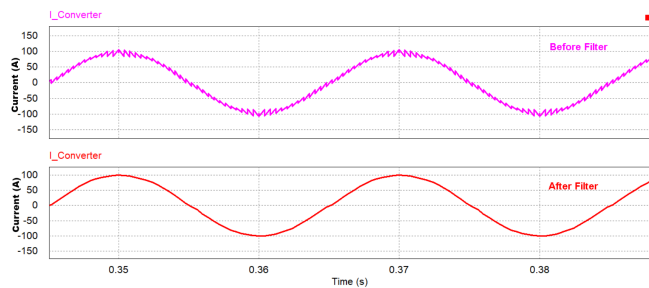


Fig. 6. Inverter current waveform before and after Filter

VI. CONCLUSION

Method 1's lower inductance value presents a range of compelling advantages for the design and operation of a 50 kVAR STATCOM. The reduced inductance allows for higher switching frequencies, which directly translate into more efficient performance. By operating at higher frequencies, the system can utilize smaller reactive components, significantly reducing the overall size and weight of the STATCOM. This compact, lightweight design is especially advantageous in applications where space is at a premium or where a more portable, flexible solution is needed.

The simulation results using the inductance values from Method 1 for the LCL filter have been exceptional, with the system producing a high-quality waveform. The lower total inductance of 350 microhenries (μH) not only facilitated higher switching frequencies but also contributed to improved system responsiveness and overall efficiency. This combination of enhanced performance, reduced size, and improved operational efficiency makes Method 1's approach highly attractive for modern STATCOM applications, offering both technical and practical benefits.

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